

Seismology Problem Set 2 – Solutions

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1. (a) Combining the two integrals,

$$\tau(q) = T - q\Delta = \int_{r_t}^b \frac{2(r^2 - \alpha^2 q^2)}{\alpha r \sqrt{r^2 - \alpha^2 q^2}} dr = \int_{r_t}^b \frac{2\sqrt{r^2 - \alpha^2 q^2}}{\alpha r} dr = 2 \int_{r_t}^b \sqrt{\eta^2 - q^2} \frac{dr}{r},$$

using $\sqrt{r^2 - \alpha^2 q^2}/\alpha = \sqrt{\eta^2 - q^2}$. Differentiating with respect to q , the lower limit r_t depends on q , but the integrand vanishes there, and so only the explicit dependence contributes:

$$\frac{d\tau}{dq} = -2 \int_{r_t}^b \frac{q}{\sqrt{\eta^2 - q^2}} \frac{dr}{r} = - \int_{r_t}^b \frac{2\alpha q}{r \sqrt{r^2 - \alpha^2 q^2}} dr = -\Delta.$$

Since $T = \tau + q\Delta$, we have $dT/dq = d\tau/dq + \Delta + q d\Delta/dq = q d\Delta/dq$, and hence $dT/d\Delta = q$. The slope of the travel-time curve at a given epicentral angle is therefore the ray parameter of the arrival, which is also, by the definition $q = r \sin \theta/\alpha$, the horizontal slowness of the wave at the receiver.

- (b) With η as the variable of integration, $dr/r = (d \ln r / d\eta) d\eta$, the turning point r_t corresponds to $\eta = q$, and the surface to $\eta = \eta_b$, so that

$$\Delta(q) = \int_{r_t}^b \frac{2q}{r \sqrt{\eta^2 - q^2}} dr = 2 \int_q^{\eta_b} \frac{q}{\sqrt{\eta^2 - q^2}} \frac{d \ln r}{d\eta} d\eta,$$

where the monotonicity of η is needed for the change of variables to be well-defined.

- (c) Multiplying by $(q^2 - \eta_1^2)^{-1/2}$ and integrating,

$$\int_{\eta_1}^{\eta_b} \frac{\Delta(q)}{\sqrt{q^2 - \eta_1^2}} dq = 2 \int_{\eta_1}^{\eta_b} dq \int_q^{\eta_b} d\eta \frac{q}{\sqrt{(\eta^2 - q^2)(q^2 - \eta_1^2)}} \frac{d \ln r}{d\eta}.$$

The region of integration is $\eta_1 \leq q \leq \eta \leq \eta_b$, so exchanging the order gives

$$2 \int_{\eta_1}^{\eta_b} d\eta \frac{d \ln r}{d\eta} \int_{\eta_1}^{\eta} \frac{q dq}{\sqrt{(\eta^2 - q^2)(q^2 - \eta_1^2)}} = 2 \cdot \frac{\pi}{2} \int_{\eta_1}^{\eta_b} \frac{d \ln r}{d\eta} d\eta = \pi \ln \frac{b}{r_1},$$

as required. To convert to an integral over the travel-time curve, note that the derivative of $\cosh^{-1}(q/\eta_1)$ with respect to q is $(q^2 - \eta_1^2)^{-1/2}$, so that integrating by parts

$$\int_{\eta_1}^{\eta_b} \frac{\Delta(q)}{\sqrt{q^2 - \eta_1^2}} dq = \left[\Delta(q) \cosh^{-1} \frac{q}{\eta_1} \right]_{\eta_1}^{\eta_b} - \int_{\eta_1}^{\eta_b} \cosh^{-1} \left(\frac{q}{\eta_1} \right) \frac{d\Delta}{dq} dq.$$

The boundary term vanishes at both ends, since $\cosh^{-1}(1) = 0$ and $\Delta(\eta_b) = 0$ (the ray of parameter η_b grazes the surface). As q increases from η_1 to η_b , Δ decreases from Δ_1 to 0, so reversing the direction of integration gives

$$\ln \frac{b}{r_1} = \frac{1}{\pi} \int_0^{\Delta_1} \cosh^{-1} \left[\frac{q(\Delta)}{\eta_1} \right] d\Delta.$$

- (d) From the observed travel-time curve we obtain $q(\Delta) = dT/d\Delta$ by differentiation. Choosing any epicentral angle Δ_1 , the ray arriving there has parameter $\eta_1 = q(\Delta_1)$, and the integral gives the radius r_1 at which it turned. But at the turning point $\eta(r_1) = \eta_1$, and hence

$\alpha(r_1) = r_1/\eta_1$. Repeating for increasing Δ_1 builds up $\alpha(r)$ from the surface downwards; note that the integral for a given depth requires the travel-time curve only out to Δ_1 , i.e. only the rays turning above that depth. If the velocity decreases with depth so rapidly that η increases with depth, then rays entering that region are bent downwards rather than turning, and no ray turns within the *low-velocity zone*. There is then a range of η with no corresponding turning radius, the change of variables in (b) fails, and the velocity within the zone cannot be recovered from the travel-time curve at all; the zone produces a gap in the curve (a shadow zone) followed by a jump in Δ , and the depths inferred below it are in error unless the zone is accounted for separately. At a discontinuity across which the velocity increases with depth, η jumps downwards and there is a range of ray parameters for which the rays are totally reflected at the interface. Rays turning just beneath it emerge at larger epicentral angles than those turning just above, so the travel-time curve folds back on itself to form a *triplication*: over a range of Δ three arrivals are seen, and $q(\Delta)$ is multi-valued. The inversion remains valid provided the branches are correctly identified, since $\Delta(q)$ itself remains single-valued.

2. From Lecture 19, we know that the least squares solution for the given problem is

$$\mathbf{m} = (\mathbf{A}^T \mathbf{C}^{-1} \mathbf{A} + \lambda \mathbf{B})^{-1} \mathbf{A}^T \mathbf{C}^{-1} \mathbf{d},$$

with $\mathbf{d}$ the data vector. Here we take $\mathbf{d} = \mathbf{A}\mathbf{m}_{\text{in}}$, and hence find that

$$\mathbf{m}_{\text{out}} = (\mathbf{A}^T \mathbf{C}^{-1} \mathbf{A} + \lambda \mathbf{B})^{-1} \mathbf{A}^T \mathbf{C}^{-1} \mathbf{A} \mathbf{m}_{\text{in}}.$$

It follows that the resolution matrix takes the form

$$\mathbf{R} = (\mathbf{A}^T \mathbf{C}^{-1} \mathbf{A} + \lambda \mathbf{B})^{-1} \mathbf{A}^T \mathbf{C}^{-1} \mathbf{A}.$$

The resolution matrix relates the input and output of a synthetic inversion that does not include data errors. In an ideal case, we would like the resulting model to be equal to the input, and hence we want $\mathbf{R}$ to be close to the identity matrix.

3. (a) The condition is that $\|\mathbf{A}\mathbf{m} - \mathbf{d}\|^2 \leq \chi_c^2$. Expanding this, we have

$$(\mathbf{A}\mathbf{m} - \mathbf{d})^T (\mathbf{A}\mathbf{m} - \mathbf{d}) = \mathbf{m}^T \mathbf{A}^T \mathbf{A} \mathbf{m} - 2\mathbf{m}^T \mathbf{A}^T \mathbf{d} + \mathbf{d}^T \mathbf{d} \leq \chi_c^2.$$

Since the number of model parameters is greater than the number of data points ($m > n$), the problem is underdetermined. This means the $n \times m$ matrix $\mathbf{A}$ necessarily has a non-trivial null space. Consequently, the $m \times m$ matrix $\mathbf{G} = \mathbf{A}^T \mathbf{A}$ is positive semi-definite but *singular* (not invertible). The singularity implies that the misfit is insensitive to any component of $\mathbf{m}$ that lies in the null space of $\mathbf{A}$. The region defined by the inequality is therefore unbounded in these null-space directions. This type of unbounded, ellipse-like shape is a *hyper-cylinder* (or more precisely, an ellipsoidal cylinder).

- (b) We want to minimise $\|\mathbf{m}\|^2$ subject to the constraint $\|\mathbf{A}\mathbf{m} - \mathbf{d}\|^2 = \chi_c^2$. The Lagrangian is

$$\mathcal{L}(\mathbf{m}, \mu) = \mathbf{m}^T \mathbf{m} + \mu (\|\mathbf{A}\mathbf{m} - \mathbf{d}\|^2 - \chi_c^2).$$

To find the minimum, we set the derivative with respect to $\mathbf{m}$ to zero:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{m}} = 2\mathbf{m} + 2\mu \mathbf{A}^T (\mathbf{A}\mathbf{m} - \mathbf{d}) = \mathbf{0}.$$

Rearranging this gives

$$(\mathbf{I} + \mu \mathbf{A}^T \mathbf{A}) \mathbf{m} = \mu \mathbf{A}^T \mathbf{d},$$

which can be solved for $\mathbf{m}$ to yield

$$\mathbf{m} = \left(\mathbf{A}^T \mathbf{A} + \frac{1}{\mu} \mathbf{I} \right)^{-1} \mathbf{A}^T \mathbf{d}.$$

The Lagrange multiplier μ is then found by substituting this expression for $\mathbf{m}$ back into the constraint equation, $\|\mathbf{A}\mathbf{m} - \mathbf{d}\|^2 = \chi_c^2$.

This result is identical in form to the *regularised least squares solution*

$$\mathbf{m}_{\text{reg}} = (\mathbf{A}^T \mathbf{A} + \lambda \mathbf{I})^{-1} \mathbf{A}^T \mathbf{d},$$

if we identify the regularisation parameter λ with $1/\mu$. The difference lies in how the regularisation parameter is chosen. In standard regularisation, λ is often chosen subjectively. In this approach, however, the parameter is determined objectively by the requirement that the misfit must equal a statistically determined value, χ_c^2 . This is not, however, to say that the resulting model is the correct one. But it does have the potentially desirable property that of all the models that fit the data it has the smallest norm.

4. A suitable Lagrangian for this problem can be defined by

$$L = J(\mathbf{u}) - \int_M \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) u'_i d^3 \mathbf{x} + \int_{\partial M} \left[\hat{n}_j \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) - \sigma g_i \right] w'_i dS,$$

where we have used the elastic tensor $A_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$, and u'_i and w'_i are Lagrange multipliers. As usual, variation with respect to the multipliers just gives the forward problem. Varying u_i , however, leads to the new condition

$$\int_{\partial M} h'_i \delta u_i dS - \int_M \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) u'_i d^3 \mathbf{x} + \int_{\partial M} \hat{n}_j \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) w'_i dS = 0,$$

where we have defined

$$\mathbf{h}'(\mathbf{x}) = \sum_{i=1}^m \frac{1}{\varepsilon_i^2} [\mathbf{u}(\mathbf{x}_i) - \mathbf{u}_i^{\text{obs}}] \delta(\mathbf{x} - \mathbf{x}_i),$$

to simplify the contribution associated with $J(\mathbf{u})$. To proceed, we need to integrate by parts to isolate δu_i . The key step is

$$\begin{aligned} & - \int_M \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) u'_i d^3 \mathbf{x} \\ &= - \int_M \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} u'_i \right) d^3 \mathbf{x} + \int_M A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \frac{\partial u'_i}{\partial x_j} d^3 \mathbf{x} \\ &= - \int_{\partial M} \hat{n}_j \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) u'_i dS + \int_M A_{klij} \frac{\partial \delta u_i}{\partial x_j} \frac{\partial u'_k}{\partial x_l} d^3 \mathbf{x} \\ &= - \int_{\partial M} \hat{n}_j \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) u'_i dS + \int_M \frac{\partial}{\partial x_j} \left(A_{ijkl} \delta u_i \frac{\partial u'_k}{\partial x_l} \right) d^3 \mathbf{x} - \int_M \delta u_i \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u'_k}{\partial x_l} \right) d^3 \mathbf{x} \\ &= - \int_{\partial M} \hat{n}_j \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) u'_i dS + \int_{\partial M} \delta u_i \hat{n}_j \left(A_{ijkl} \frac{\partial u'_k}{\partial x_l} \right) dS - \int_M \delta u_i \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u'_k}{\partial x_l} \right) d^3 \mathbf{x}, \end{aligned}$$

where we note that the hyperelastic symmetry $A_{ijkl} = A_{klij}$ has been used. Using this result, the condition now reads

$$\int_{\partial M} [h'_i + \hat{n}_j \left(A_{ijkl} \frac{\partial u'_k}{\partial x_l} \right)] \delta u_i dS - \int_M \delta u_i \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u'_k}{\partial x_l} \right) d^3 \mathbf{x} + \int_{\partial M} \hat{n}_j \left(A_{ijkl} \frac{\partial \delta u_k}{\partial x_l} \right) (w'_i - u'_i) dS = 0.$$

It follows that u'_i must satisfy

$$\frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u'_k}{\partial x_l} \right) = 0,$$

subject to the boundary condition

$$\hat{n}_j \left(A_{ijkl} \frac{\partial u'_k}{\partial x_l} \right) = -h'_i,$$

while also $w'_i = u'_i$ on ∂M . These are the adjoint equations for this problem. Having solved the forward and adjoint problems, the first-order perturbation to J can be written

$$\delta J = - \int_M \frac{\partial}{\partial x_j} \left(\delta A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) u'_i d^3 \mathbf{x} + \int_{\partial M} \left[\hat{n}_j \left(\delta A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) - \delta \sigma g_i \right] u'_i dS.$$

Performing an integration by parts, this simplifies to

$$\delta J = \int_M \delta A_{ijkl} \frac{\partial u_k}{\partial x_l} \frac{\partial u'_i}{\partial x_j} d^3 \mathbf{x} - \int_{\partial M} \delta \sigma g_i u'_i dS.$$

From this we see that the sensitivity kernel with respect to σ is equal to $K_\sigma = -g_i u'_i$. To get the kernels for λ and μ , we note that

$$\delta A_{ijkl} = \delta \lambda \delta_{ij} \delta_{kl} + \delta \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}),$$

from which we find

$$\delta J = \int_M \delta \lambda \frac{\partial u_i}{\partial x_i} \frac{\partial u'_j}{\partial x_j} d^3 \mathbf{x} + \frac{1}{2} \int_M \delta \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right) d^3 \mathbf{x},$$

where we have set $\delta \sigma = 0$ as that term has been dealt with. From this expression, we can now read off the desired sensitivity kernels

$$K_\lambda = \frac{\partial u_i}{\partial x_i} \frac{\partial u'_j}{\partial x_j},$$

$$K_\mu = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \left(\frac{\partial u'_i}{\partial x_j} + \frac{\partial u'_j}{\partial x_i} \right).$$

As is typical with the adjoint method, we see that these sensitivity kernels depend on both the solution of the forward and adjoint problems. These two problems are identical in form, differing only in their force terms, and hence the overall cost of the calculations is that of two forward calculations.

5. The part of the Lagrangian that depends explicitly on the motion is

$$\frac{1}{2} G \int_M \frac{\rho \rho'}{\{[\varphi_i - \varphi'_i][\varphi_i - \varphi'_i]\}^{\frac{1}{2}}} d^3 \mathbf{x}',$$

where it is understood that unprimed functions depend on $\mathbf{x}$ and primed ones on the dummy integration variable $\mathbf{x}'$. Perturbing the motion to $\varphi + \delta \varphi$ and expanding to first-order accuracy we find

$$\int_M \frac{\delta \mathcal{L}}{\delta \varphi_i} \delta \varphi_i d^3 \mathbf{x} = -\frac{1}{2} G \int_M \int_M \frac{\rho \rho' (\varphi_i - \varphi'_i) (\delta \varphi_i - \delta \varphi'_i)}{\{[\varphi_j - \varphi'_j][\varphi_j - \varphi'_j]\}^{\frac{3}{2}}} d^3 \mathbf{x}' d^3 \mathbf{x}.$$

Splitting this expression into two parts we find

$$\begin{aligned} \int_M \frac{\delta \mathcal{L}}{\delta \varphi_i} \delta \varphi_i d^3 \mathbf{x} &= -\frac{1}{2} G \int_M \int_M \frac{\rho \rho' (\varphi_i - \varphi'_i) \delta \varphi_i}{\{[\varphi_j - \varphi'_j][\varphi_j - \varphi'_j]\}^{\frac{3}{2}}} d^3 \mathbf{x}' d^3 \mathbf{x} \\ &\quad + \frac{1}{2} G \int_M \int_M \frac{\rho \rho' (\varphi_i - \varphi'_i) \delta \varphi'_i}{\{[\varphi_j - \varphi'_j][\varphi_j - \varphi'_j]\}^{\frac{3}{2}}} d^3 \mathbf{x}' d^3 \mathbf{x}, \end{aligned}$$

but by interchanging the order of integration and relabelling dummy variables $\mathbf{x} \leftrightarrow \mathbf{x}'$, we note that the two terms are actually equal, and hence

$$\int_M \frac{\delta \mathcal{L}}{\delta \varphi_i} \delta \varphi_i d^3 \mathbf{x} = -G \int_M \delta \varphi_i \left(\int_M \frac{\rho \rho' (\varphi_i - \varphi'_i)}{\{[\varphi_j - \varphi'_j][\varphi_j - \varphi'_j]\}^{\frac{3}{2}}} d^3 \mathbf{x}' \right) d^3 \mathbf{x},$$

which implies

$$\frac{\delta \mathcal{L}}{\delta \varphi_i} = -G \rho \int_M \frac{\rho' (\varphi_i - \varphi'_i)}{\{[\varphi_j - \varphi'_j][\varphi_j - \varphi'_j]\}^{\frac{3}{2}}} d^3 \mathbf{x}' = \rho \gamma_i,$$

as required.

6. (a) Expanding about the source point,

$$\frac{\partial |km\rangle^*}{\partial x'_j}(\mathbf{x}') = \frac{\partial |km\rangle^*}{\partial x_j}(\mathbf{x}_s) + (x'_l - x_{s,l}) \frac{\partial^2 |km\rangle^*}{\partial x_j \partial x_l}(\mathbf{x}_s) + \dots,$$

and so the spatial integral in the mode sum becomes

$$\begin{aligned} \int_M \bar{S}_{ij}(\mathbf{x}', t') \frac{\partial |km\rangle^*}{\partial x'_j}(\mathbf{x}') d^3\mathbf{x}' &= M_{ij}(t') \frac{\partial |km\rangle^*}{\partial x_j}(\mathbf{x}_s) \\ &+ \frac{\partial^2 |km\rangle^*}{\partial x_j \partial x_l}(\mathbf{x}_s) \int_M (x'_l - x_{s,l}) \bar{S}_{ij}(\mathbf{x}', t') d^3\mathbf{x}' + \dots \end{aligned}$$

The first term is exactly what is obtained from the point source with moment tensor $M_{ij}(t)$, and the second is smaller than it by a factor of order L/λ_k , where λ_k is the length over which the strain of the eigenfunction changes appreciably. The eigenfunction of a mode with frequency ω_k oscillates in space with a wavelength of order $\lambda_k \sim 2\pi v/\omega_k$, with v a typical wave speed, so the point-source approximation requires $L \ll v/f_k$ where $f_k = \omega_k/2\pi$. Since the seismogram is a sum over modes, the approximation is good at frequencies for which $f \ll v/L$, and fails at higher frequencies whatever the size of the fault; it is therefore a statement about the frequency band retained. For $L = 100$ km and $v = 5$ km s⁻¹, $v/L = 50$ mHz, so the approximation is reasonable below perhaps 10 mHz (periods above about 100 s, wavelengths above 500 km) and certainly for the free oscillations discussed in the lectures.

- (b) We are told

$$\mathbf{u}(\mathbf{x}, t) = \sum_{km} \left\{ \int_0^t \frac{\sin[\omega_k(t-t')]}{\omega_k} \int_M \bar{S}_{ij}(\mathbf{x}', t') \frac{\partial |km\rangle^*}{\partial x'_j}(\mathbf{x}') d^3\mathbf{x}' dt' \right\} |km\rangle(\mathbf{x}),$$

noting that arguments have now been included for clarity. In the case of a moment tensor point source, we see immediately that

$$\int_M \bar{S}_{ij}(\mathbf{x}', t') \frac{\partial |km\rangle^*}{\partial x'_j}(\mathbf{x}') d^3\mathbf{x}' = M_{ij} \frac{\partial |km\rangle^*}{\partial x_j}(\mathbf{x}_s) H(t' - t_s),$$

while the necessary time integral is readily evaluated for $t \geq t_s$:

$$\begin{aligned} \int_0^t \frac{\sin[\omega_k(t-t')]}{\omega_k} H(t' - t_s) dt' &= \int_{t_s}^t \frac{\sin[\omega_k(t-t')]}{\omega_k} dt' \\ &= \left[\frac{\cos[\omega_k(t-t')]}{\omega_k^2} \right]_{t_s}^t = \frac{1 - \cos[\omega_k(t-t_s)]}{\omega_k^2}, \end{aligned}$$

and hence the eigenfunction expansion reduces to

$$\mathbf{u}(\mathbf{x}, t) = \sum_{km} \frac{1 - \cos[\omega_k(t-t_s)]}{\omega_k^2} M_{ij} \frac{\partial |km\rangle^*}{\partial x_j}(\mathbf{x}_s) |km\rangle(\mathbf{x}),$$

for $t \geq t_s$ and zero before this. This final expression depends on ten source parameters, assuming that Earth structure is known. The dependence on the moment tensor is linear, but that on the source location and source time is non-linear. There are several possible methods for solving the inverse problem. The most obvious is to define some sort of least-squares misfit between observed and predicted seismograms, and then optimise to find the best-fitting source parameters. To do this gradient-based methods could be applied, and here the derivatives with respect to the source parameters could be calculated directly with relative ease. Note that in this problem there are likely to be more data than model parameters, and hence it is likely to be overdetermined. As a result, regularisation would not be needed. Because the problem is non-linear, it is not certain that the optimal solution obtained would be a global minimum. Due to the size of the inverse problem and the low cost of implementing mode summation (in a spherically symmetric earth model at least) the application of Bayesian methods via something like MCMC would be feasible, and this could address the possible issue of local minima, while also providing a quantification of uncertainty.

7. Within the eigenvalue problem we take the test function equal to $\mathbf{s}$ and so obtain

$$-\omega^2 + i\omega\langle\mathbf{s}|W|\mathbf{s}\rangle + \langle\mathbf{s}|H|\mathbf{s}\rangle = 0,$$

where we have used the normalisation condition. This is a quadratic equation for ω , and its solution yields

$$\omega = \frac{1}{2}i\langle\mathbf{s}|W|\mathbf{s}\rangle \pm \sqrt{\langle\mathbf{s}|H|\mathbf{s}\rangle - \frac{1}{4}\langle\mathbf{s}|W|\mathbf{s}\rangle^2},$$

as required. Note that the Coriolis operator is anti-Hermitian, and hence $\langle\mathbf{s}|W|\mathbf{s}\rangle$ is an imaginary number. For this reason the expression for ω is real so long as

$$\langle\mathbf{s}|H|\mathbf{s}\rangle - \frac{1}{4}\langle\mathbf{s}|W|\mathbf{s}\rangle^2 > 0,$$

with the contribution from the Coriolis term always being non-negative. We see that in the rotating eigenvalue problem the Coriolis force always contributes to the stability of the earth model. As with the non-rotating case, the question of stability reduces to the properties of the potential energy form which here includes an additional contribution from centrifugal forces. This latter part can be written explicitly as

$$\int_M \rho \epsilon_{ijk} \epsilon_{klm} s_i^* \Omega_j \Omega_l s_m d^3\mathbf{x},$$

and using the identity $\epsilon_{ijk}\epsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$ this becomes

$$\int_M \rho \epsilon_{ijk} \epsilon_{klm} s_i^* \Omega_j \Omega_l s_m d^3\mathbf{x} = \int_M \rho [|\mathbf{s} \cdot \boldsymbol{\Omega}|^2 - \|\mathbf{s}\|^2 \|\boldsymbol{\Omega}\|^2] d^3\mathbf{x}.$$

Applying the Cauchy–Schwarz inequality we note that this term is always non-positive, and hence centrifugal forces are destabilising.

8. (a) Build up the planet by bringing in successive spherical shells from infinity. When the radius is r , the mass already assembled is $m(r) = \frac{4}{3}\pi\rho r^3$, and adding a shell of mass $dm = 4\pi\rho r^2 dr$ releases energy $Gm dm/r$. Thus

$$E_g = - \int_0^a \frac{Gm(r)}{r} 4\pi\rho r^2 dr = -\frac{16\pi^2}{3} G\rho^2 \int_0^a r^4 dr = -\frac{16\pi^2}{15} G\rho^2 a^5 = -\frac{3GM^2}{5a},$$

where $M = \frac{4}{3}\pi\rho a^3$.

- (b) Within a uniform sphere the gravitational acceleration is $g(r) = \frac{4}{3}\pi G\rho r$ directed inwards, and the hydrostatic equation $dp_0/dr = -\rho g$ with $p_0(a) = 0$ integrates to the given expression. Then

$$3 \int_M p_0 d^3\mathbf{x} = 8\pi^2 G\rho^2 \int_0^a (a^2 - r^2)r^2 dr = 8\pi^2 G\rho^2 \left(\frac{a^5}{3} - \frac{a^5}{5} \right) = \frac{16\pi^2}{15} G\rho^2 a^5 = -E_g.$$

This is a form of the virial theorem for a body in hydrostatic equilibrium.

- (c) The mass is unchanged while the radius becomes $a(1 - \epsilon)$, so

$$\Delta E_g = -\frac{3GM^2}{5a} \left[\frac{1}{1 - \epsilon} - 1 \right] = -\frac{3GM^2}{5a} (\epsilon + \epsilon^2) + O(\epsilon^3).$$

As expected, contraction releases gravitational energy.

- (d) Under the uniform compression every material element has its linear dimensions scaled by $1 - \epsilon$, so its fractional decrease in volume is $s = 1 - (1 - \epsilon)^3 = 3\epsilon - 3\epsilon^2 + O(\epsilon^3)$. The work done on an element of initial volume V_0 is

$$\int_0^s (p_0 + \kappa s') V_0 ds' = V_0 \left(p_0 s + \frac{1}{2} \kappa s^2 \right) = V_0 \left[(3\epsilon - 3\epsilon^2) p_0 + \frac{9}{2} \kappa \epsilon^2 \right] + O(\epsilon^3),$$

and summing over the planet gives the stated result. Note that because the compression is uniform there is no shear strain, and so the shear modulus plays no role.

(e) Adding the two contributions, the first-order terms are

$$3\epsilon \int_M p_0 d^3\mathbf{x} - \frac{3GM^2}{5a}\epsilon = \epsilon \left[3 \int_M p_0 d^3\mathbf{x} + E_g \right] = 0,$$

by part (b). This had to be so because the initial state is an equilibrium, and hence a stationary point of the total energy. Using $\int_M p_0 d^3\mathbf{x} = -E_g/3 = GM^2/5a$, the second-order terms are

$$\Delta E = \epsilon^2 \left[\frac{9}{2}\kappa V - 3 \int_M p_0 d^3\mathbf{x} - \frac{3GM^2}{5a} \right] = \epsilon^2 \left[\frac{9}{2}\kappa V - \frac{6GM^2}{5a} \right] = \frac{9}{2}V\epsilon^2 \left[\kappa - \frac{4}{3}\bar{p}_0 \right],$$

where $\bar{p}_0 = \frac{1}{V} \int_M p_0 d^3\mathbf{x} = GM^2/5aV$. Stability with respect to uniform compression therefore requires $\kappa > \frac{4}{3}\bar{p}_0$; the second-order gravitational term alone would be balanced by an elastic energy twice as small, and it is the release of energy stored in the hydrostatic pressure that accounts for the rest. Using $M = \frac{4}{3}\pi\rho a^3$ and $V = \frac{4}{3}\pi a^3$ we find $\bar{p}_0 = \frac{4\pi}{15}G\rho^2 a^2$, and hence $\kappa > \frac{16\pi}{45}G\rho^2 a^2$. The surface gravity is $g = \frac{4}{3}\pi G\rho a$, so this can be written as

$$\frac{\kappa}{\rho} > \frac{4}{15}ga,$$

or, in words, the sound speed must exceed $\sqrt{4ga/15}$, a fixed fraction of the orbital speed $\sqrt{ga}$ at the surface. For the given numbers $\frac{16\pi}{45}G\rho^2 a^2 \approx 92 \text{ GPa}$, or equivalently a sound speed of about 4.1 km s^{-1} . Within PREM the bulk modulus in the mantle (and still more so in the core) lies comfortably above this value, so we should expect the Earth to be stable to a global compression, though the margin is far smaller than one might have guessed. If the material instead obeyed an adiabatic law $p \propto \rho^\gamma$, then $\kappa = \gamma p_0$, and the same argument gives the classical condition $\gamma > 4/3$ for the stability of a self-gravitating gas sphere. The calculation is only indicative: uniform compression is not an eigenfunction of the problem, and the true condition for stability requires the potential energy form of Lecture 23 to be positive for all admissible displacements.

9. (a) With $P_1 = H_1 = 0$, the first-order splitting problem of Lecture 24 reduces to

$$\frac{1}{2\omega_k} \sum_{m=-l}^l i\omega_k \langle km' | W | km \rangle a_m = \delta\omega a_{m'},$$

and substituting the given matrix elements,

$$\frac{i}{2}(-2im'\Omega\beta)a_{m'} = m'\Omega\beta a_{m'} = \delta\omega a_{m'}.$$

The matrix is diagonal, so its eigenvectors are the singlets themselves, $a_m = \delta_{mm_0}$, with eigenvalues $\delta\omega = m_0\Omega\beta$. Hence $\omega_m = \omega_k + m\Omega\beta$ for $-l \leq m \leq l$: the multiplet is split into $2l+1$ singlets with uniform spacing $\Omega\beta$, symmetric about the degenerate frequency. Their sum, $\sum_m m\Omega\beta = 0$, vanishes as the diagonal sum rule requires. In question 7 we found, for a normalised eigenfunction, $\omega = \frac{1}{2}i\langle \mathbf{s} | W | \mathbf{s} \rangle \pm \sqrt{\langle \mathbf{s} | H | \mathbf{s} \rangle - \frac{1}{4}\langle \mathbf{s} | W | \mathbf{s} \rangle^2}$. Taking $\mathbf{s} = |km\rangle$, so that $\langle \mathbf{s} | H | \mathbf{s} \rangle = \omega_k^2$ and $\langle \mathbf{s} | W | \mathbf{s} \rangle = -2im\Omega\beta$, this gives

$$\omega = m\Omega\beta + \sqrt{\omega_k^2 + m^2\Omega^2\beta^2} = \omega_k + m\Omega\beta + O(\Omega^2),$$

in agreement with the first-order result.

(b) We need $\langle km | W | km \rangle = \int_M 2\rho \mathbf{s}^* \cdot (\boldsymbol{\Omega} \times \mathbf{s}) d^3\mathbf{x}$ with $\mathbf{s} = W\mathbf{C}_{lm}$. Writing Y for Y_{lm} , and using $\hat{\mathbf{r}} \times \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\varphi}}$ and $\hat{\mathbf{r}} \times \hat{\boldsymbol{\varphi}} = -\hat{\boldsymbol{\theta}}$,

$$\mathbf{C}_{lm} = \hat{\mathbf{r}} \times \nabla_1 Y = \partial_\theta Y \hat{\boldsymbol{\varphi}} - \frac{im}{\sin\theta} Y \hat{\boldsymbol{\theta}}.$$

With $\hat{\mathbf{z}} = \cos\theta \hat{\mathbf{r}} - \sin\theta \hat{\boldsymbol{\theta}}$ we have $\hat{\mathbf{z}} \times \hat{\boldsymbol{\theta}} = \cos\theta \hat{\boldsymbol{\varphi}}$ and $\hat{\mathbf{z}} \times \hat{\boldsymbol{\varphi}} = -\cos\theta \hat{\boldsymbol{\theta}} - \sin\theta \hat{\mathbf{r}}$, so that

$$\hat{\mathbf{z}} \times \mathbf{C}_{lm} = -\partial_\theta Y (\cos\theta \hat{\boldsymbol{\theta}} + \sin\theta \hat{\mathbf{r}}) - \frac{im \cos\theta}{\sin\theta} Y \hat{\boldsymbol{\varphi}},$$

and therefore

$$\mathbf{C}_{lm}^* \cdot (\hat{\mathbf{z}} \times \mathbf{C}_{lm}) = -\frac{im \cos \theta}{\sin \theta} (Y \partial_\theta Y^* + Y^* \partial_\theta Y) = -\frac{im \cos \theta}{\sin \theta} \partial_\theta |Y|^2.$$

Integrating over the unit sphere,

$$\begin{aligned} \int_{S^2} \mathbf{C}_{lm}^* \cdot (\hat{\mathbf{z}} \times \mathbf{C}_{lm}) dS &= -im \int_0^{2\pi} \int_0^\pi \cos \theta \partial_\theta |Y|^2 d\theta d\varphi \\ &= -im \left\{ 2\pi \left[\cos \theta |Y|^2 \right]_0^\pi + \int_{S^2} |Y|^2 dS \right\} = -im, \end{aligned}$$

the boundary term vanishing because Y vanishes at the poles for $m \neq 0$, and the remaining integral being unity by orthonormality. Hence

$$\langle km | W | km \rangle = 2\Omega \int_0^b \rho W^2 r^2 dr \int_{S^2} \mathbf{C}_{lm}^* \cdot (\hat{\mathbf{z}} \times \mathbf{C}_{lm}) dS = -2im\Omega \int_0^b \rho W^2 r^2 dr = -\frac{2im\Omega}{l(l+1)},$$

using the stated normalisation, and so $\beta = 1/[l(l+1)]$. (The off-diagonal elements $m' \neq m$ vanish because the integrand is then proportional to $e^{i(m-m')\varphi}$, which integrates to zero.) A check on the spheroidal formula quoted in the question is that the same manipulations applied to $\mathbf{s} = U Y \hat{\mathbf{r}} + V \nabla_1 Y$ give the angular integral $-im(2UV + V^2)$.

- (c) With $\Omega = 2\pi/86164 \text{ s} = 7.29 \times 10^{-5} \text{ s}^{-1}$, the rotation frequency is $\Omega/2\pi = 11.6 \mu\text{Hz}$, and the spacing between adjacent singlets is $\beta\Omega/2\pi$: $1.9 \mu\text{Hz}$ for ${}_0T_2$ ($\beta = 1/6$), $4.6 \mu\text{Hz}$ for ${}_0S_2$ and $2.2 \mu\text{Hz}$ for ${}_0S_3$. The observed spectra show ${}_0S_2$ singlets separated by about $4.6\text{--}4.8 \mu\text{Hz}$ and ${}_0S_3$ singlets by about $2.2 \mu\text{Hz}$, so that the splitting of these gravest modes is almost entirely due to rotation. The spacing is not exactly uniform because the neglected effects, the Earth's ellipticity (and the centrifugal potential) and its lateral heterogeneity, also split the multiplet; to first order these contribute terms even in m and so shift the inner and outer pairs of singlets differently, and second-order effects of rotation do the same. The peaks are also broadened by attenuation and by the finite length of the record, which limits the precision with which their positions can be read off.